

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Show that Fermat's two methods of determining a maximum or minimum of a polynomial $p(x)$ are both equivalent to solving $p'(x) = 0$.

2. Use one of Fermat's methods to find the maximum of $bx - x^3$.

3. Modify Fermat's tangent method to be able to apply it to curves given by equations of the form $f(x, y) = c$. Begin by noting that if $(x + e, \bar{y})$ is a point on the tangent line near to (x, y) , then $\bar{y} = \frac{t+e}{t}y$. Then adequate $f(x, y)$ to $f(x + e, \frac{t+e}{t}y)$. A

4. Show in modern notation, Fermat's method of finding the subtangent t to $y = f(x)$ determines t as $t = f(x)/f'(x)$.